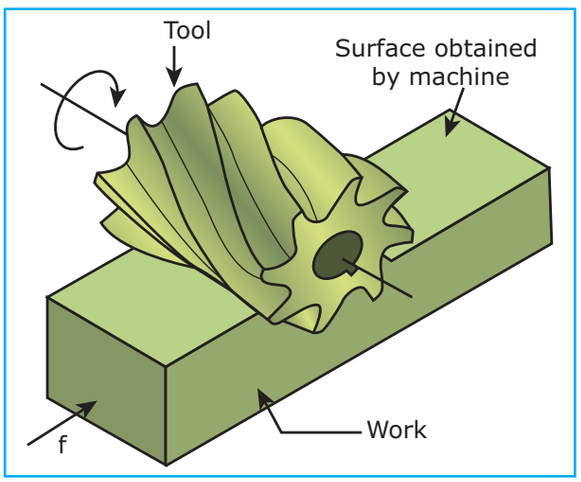
Milling machine was invented by Eli Whitney in 1818.

In 1861, it was developed by Brown & sharp, Joseph R. Brown to provide various special features of the machine.



5.1 Introduction

Milling is a process of removing metal by feeding the work against a rotating multipoint cutter. The machine tool intended for this purpose is known as milling machine.



It is found in shops where tools and cutters are manufactured. The surface obtained by this machine tool is superior in quality and more accurate and precise. Eli Whitney designed a complete milling machine in 1818. In the year 1861 Joseph Brown, a member of Brown and Sharp Company developed the first universal milling machine.

Advantages

1. The metal is removed at a faster rate as the cutter has got multiple cutting edges and rotates at a higher speed.
2. It is possible to perform machining by mounting more than one cutter at a time.
3. The able of the machine can be moved to an accuracy of 0.02 mm.
4. It is very useful since various cutters and precise tools can be machined.
5. Special attachments can be mounted on the machine to perform operations that are performed in other machine tools.

Disadvantages

1. The cost of the milling machine is high.
2. As milling cutters cost high, the investment for procuring tools is more.
3. The production cost will increase if we carry out the operations performed in a shaper or a drilling machine with a milling machine.

5.2 Horizontal Milling Machine

Base

It is made of cast iron and supports all the other parts of the machine tool. A vertical column is mounted upon the base. In some machines, the base serves as a reservoir for cutting fluid.

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